

DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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CLI To-Do List Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

This project defines a data model for a CLI-based To-Do List Manager. It implements a persistent storage pattern using a local JSON file (`~/ . todos . json`) to manage task entities. The system focuses on strict field validation, sequential ID generation, and a deterministic state machine for task lifecycle transitions (pending, completed, deleted).

1.2 Maturity Assessment

The specification provides a solid foundation for the data schema but is currently in REFINEMENT. While the field definitions are precise, the project lacks critical high-severity components including explicit Acceptance Criteria and a Testing & Validation strategy for edge cases like JSON corruption and ID collisions.

1.3 Technical Stack

- **Storage Format:** JSON
- **Timestamp Standard:** ISO 8601 UTC
- **Storage Path:** `~/ . todos . json`

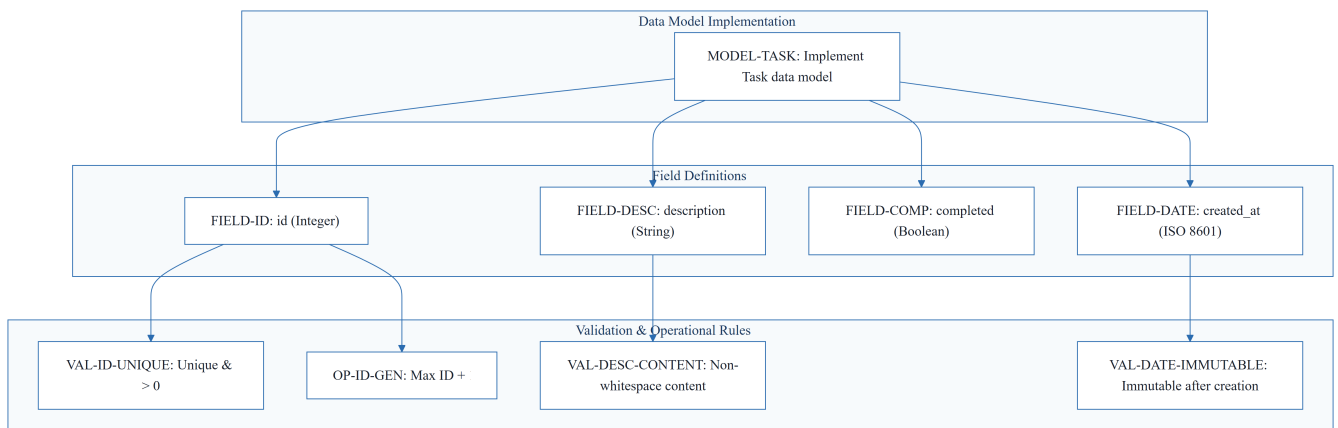
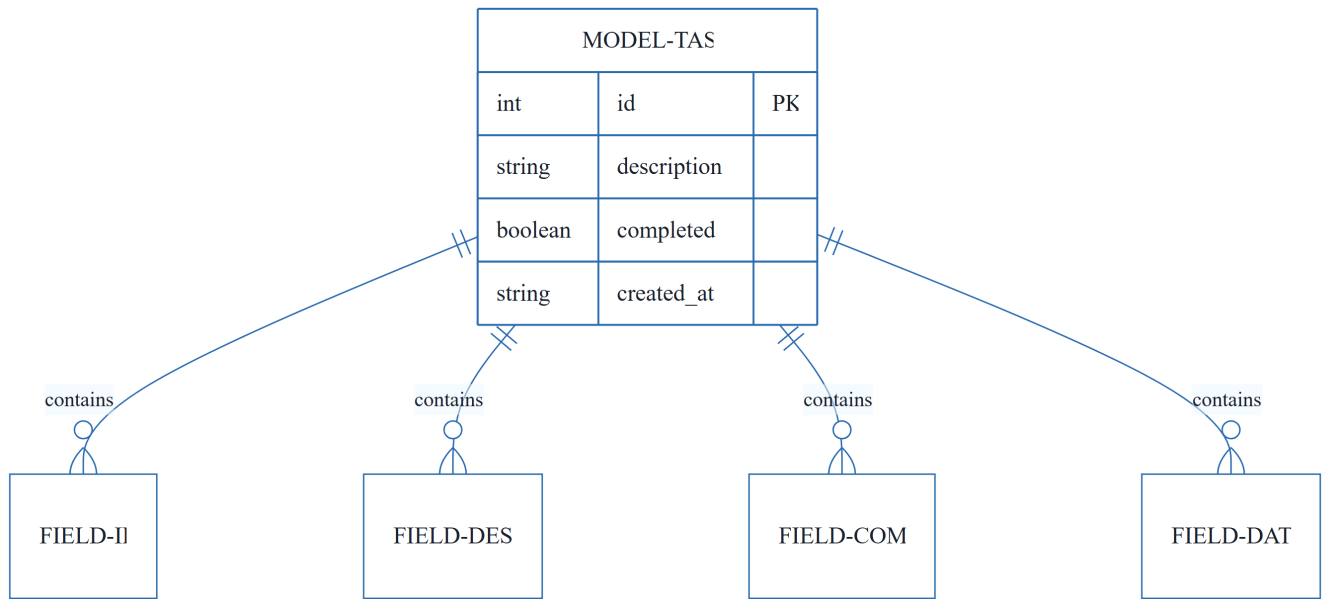
1.4 Architectural Constraints

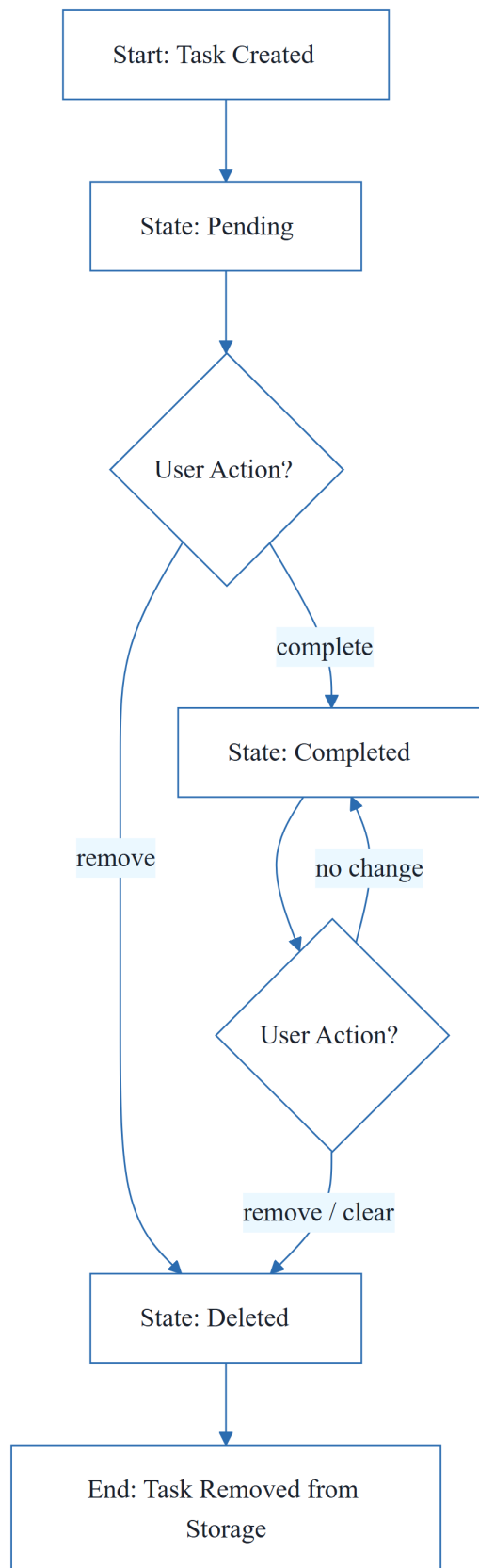
- Storage location fixed to `~/ . todos . json`.
- ID generation must follow the "highest current ID + 1" logic.
- `created_at` timestamps must be immutable after the initial write operation.
- Corrupted JSON files must trigger a user-facing error rather than a silent failure.
- Missing storage files must be interpreted as an empty task list.

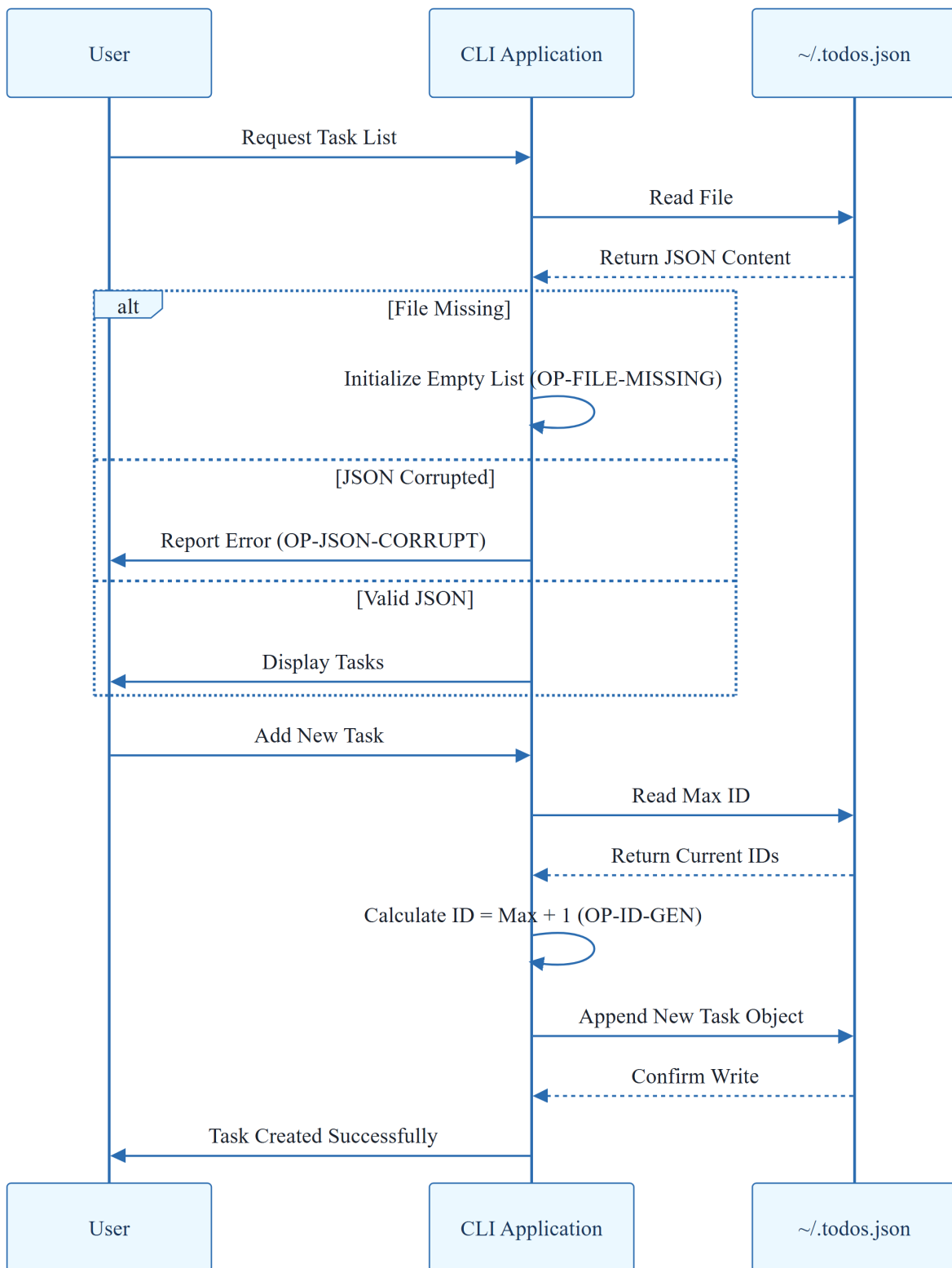
1.5 Critical Dependencies

- Local File System API for `~/ . todos . json` access.
- JSON Parser/Serializer for data persistence.
- ISO 8601 compliant timestamp generator.
- State transition dependency: State transitions depend on the underlying Task data model.

2. Architecture Workflows

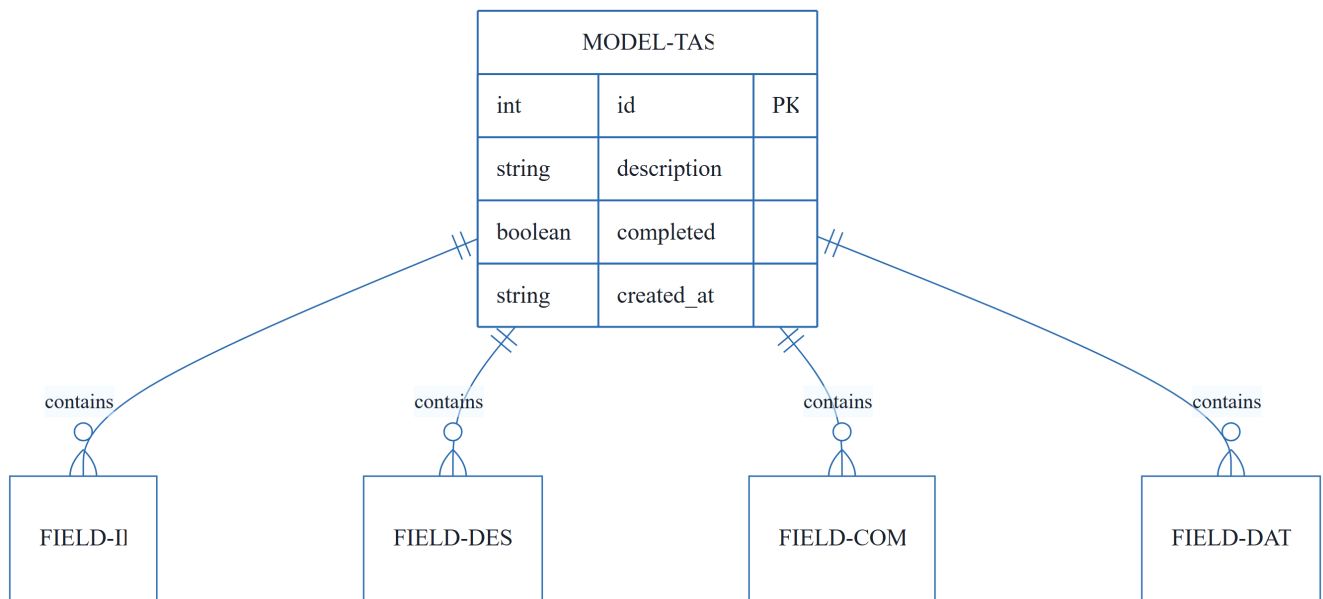




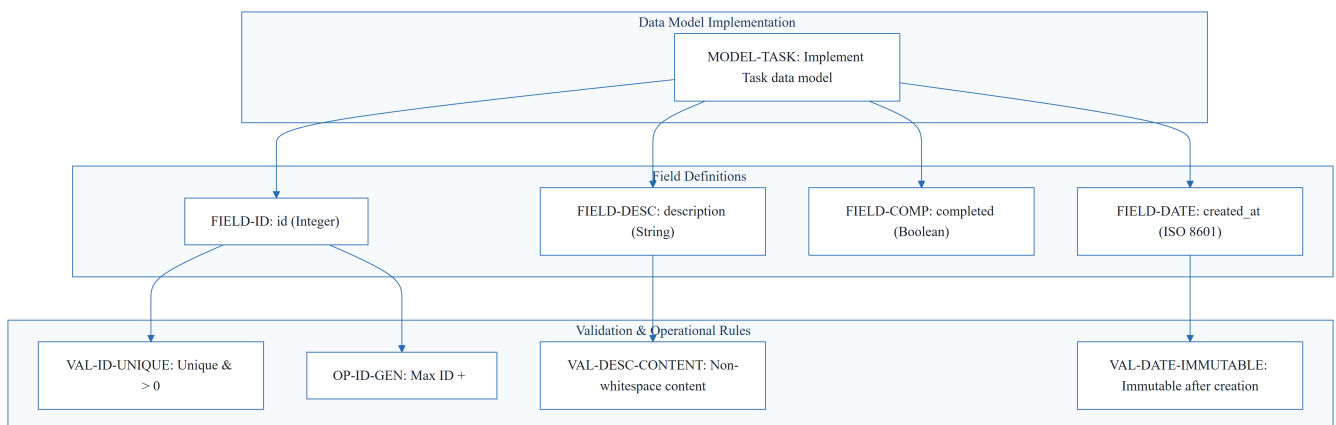


& Visual Diagrams

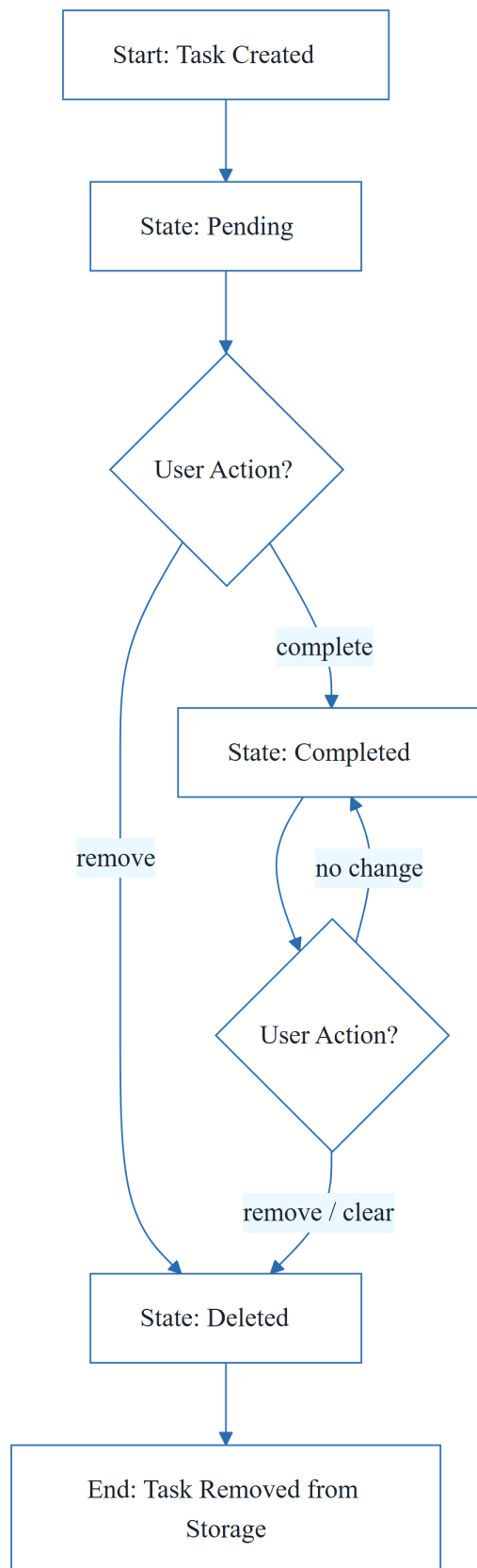
2.1 Task Data Model ER Diagram



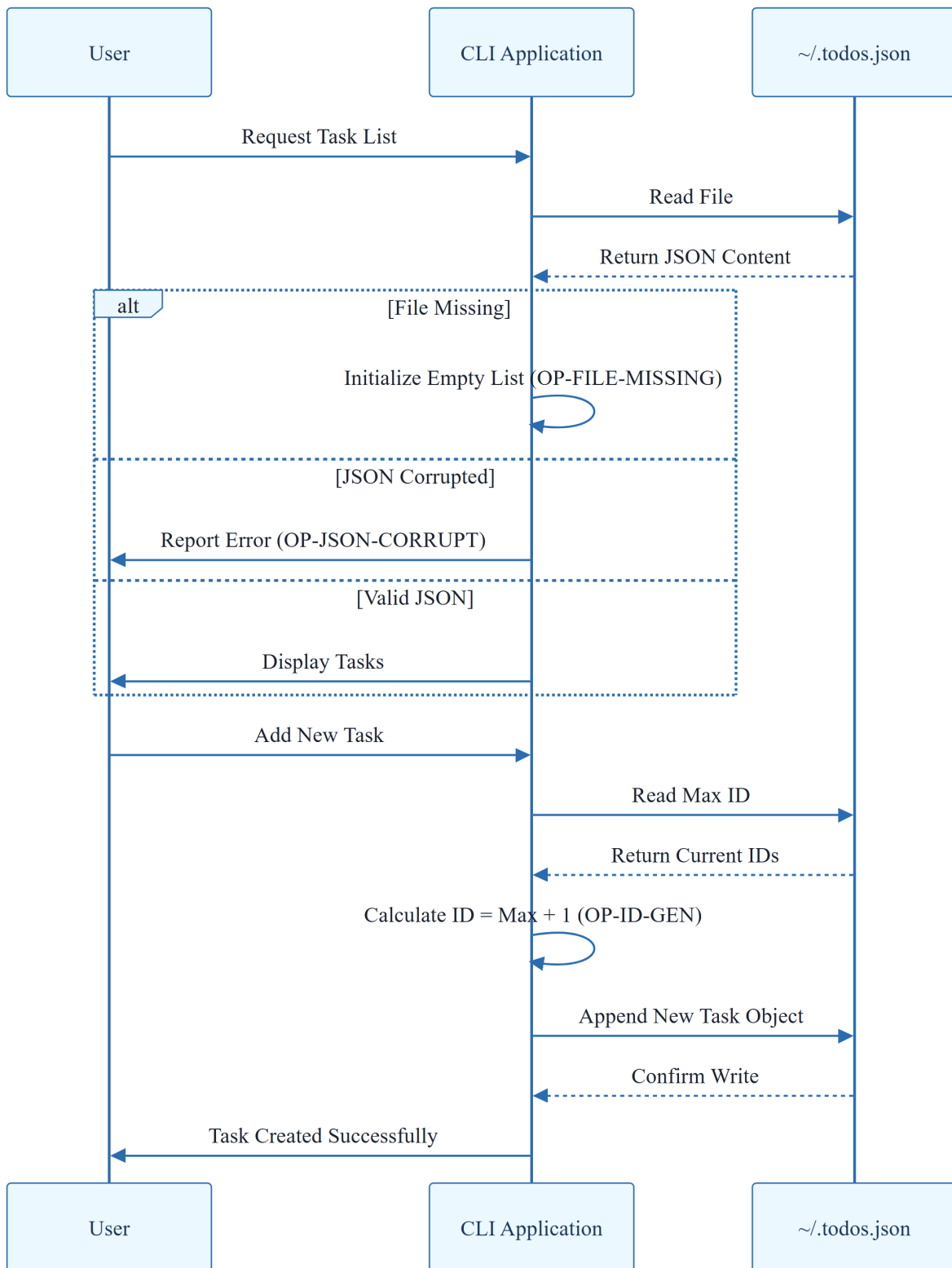
2.2 Data Model Implementation Traceability



2.3 Task State Transition Workflow



2.4 Storage Access Sequence



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

Identifier	Requirement / Component	Description	Source Section
MODEL-TASK	Task Data Model	Implement the Task data model stored in <code>~/todos.json</code>	Task
FIELD-ID	Field: id	Implement id: Integer, unique, sequential, positive, required	Fields
FIELD-DESC	Field: description	Implement description: String, required, trimmed, non-empty	Fields
FIELD-COMP	Field: completed	Implement completed: Boolean, required, default false	Fields
FIELD-DATE	Field: created_at	Implement created_at: ISO 8601 UTC timestamp, required, immutable	Fields
VAL-ID-UNIQUE	Validation: ID Unique	id must be > 0 and must not duplicate existing IDs	Validation Rules
VAL-DESC-CONTENT	Validation: Description	description must contain non-whitespace content after trimming	Validation Rules
VAL-DATE-IMMUTABLE	Validation: Date Immutable	created_at must be present at write and immutable during updates	Validation Rules
OP-FILE-MISSING	Operational: File Missing	Missing storage file must be treated as an empty task list	Operational Rules
OP-JSON-CORRUPT	Operational: JSON Corrupt	Corrupted JSON must be reported as an error to the user	Operational Rules
OP-ID-GEN	Operational: ID Generation	IDs assigned from highest current ID + 1	Operational Rules
STATE-TRANSITION	State Transitions	Implement state transitions: pending->completed, pending->deleted, completed->deleted	State Transitions

3.2 Security Rules

- **Data Integrity:** All writes to `~/todos.json` must ensure the JSON structure remains valid.
- **Input Sanitization:** The `description` field must be trimmed of leading and trailing whitespace to prevent empty-string entries.
- **Immutability:** The `created_at` field must be protected against updates once the record is persisted.

3.3 Data Models

Entity: Task

- id (Integer): Primary Key, Unique, Positive.
- description (String): Required, Non-empty.
- completed (Boolean): Default false.
- created_at (String): ISO 8601 UTC Timestamp.

Storage Format Example:

```
[
  {
    "id": 1,
    "description": "Buy groceries",
    "completed": false,
    "created_at": "2026-08-04T12:00:00Z"
  }
]
```

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Dependencies & Integration Points	LOW	Define if this model depends on specific JSON libraries or OS file system APIs.
Acceptance Criteria	HIGH	Define explicit testable criteria for the data model (e.g., 'Verify that a task cannot be saved with an empty description').
Checkboxes Checklist	MEDIUM	Create a checklist for the implementation of each field and validation rule.
Testing & Validation	HIGH	Define test cases for corrupted JSON and ID collision scenarios.
Open Questions & Uncertainties	LOW	Check if there are limits on the number of tasks or file size.

4.2 Remediation & Workflow

The project is currently in the **Refinement** phase. The immediate priority is to address the **HIGH** priority gaps by defining a comprehensive Testing & Validation suite and explicit Acceptance Criteria to ensure the data model's robustness against corruption and invalid inputs.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
ID	BUSINESS_DOMAIN	FIELD-ID	A positive, sequential, and unique integer used to distinguish each entry, calculated as the maximum existing value plus one.
JSON	TECHNICAL_STACK	Storage Model	The structured text format used for the persistent storage file located at ~/.todos.json, containing an array of objects.
UTC	TECHNICAL_STACK	FIELD-DATE	The time standard used for the immutable creation timestamp, formatted according to ISO 8601.
complete \	BUSINESS_DOMAIN	STATE-TRANSITION	The operational command that triggers a state change from pending to a finished status for a specific record.
remove \	BUSINESS_DOMAIN	STATE-TRANSITION	The operational command that transitions a record from either pending or finished status to a deleted state.